

PIYUSH KUMAR SRIVASTAVA

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SUMMARY

Mechanical Engineering undergraduate with applied Machine Learning experience across three end-to-end projects (fault classification, process-efficiency prediction, AI product development) and two manufacturing internships. Skilled across the full ML lifecycle — data cleaning, feature engineering, model building/evaluation, and deployment — using Python, SQL, Pandas, and Scikit-learn. Targeting Data Science / Machine Learning Engineer roles.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL), R

Machine Learning: Scikit-learn, TensorFlow, Regression, Classification, SVM, Model Evaluation, Predictive Modeling

Data & Analysis: Pandas, NumPy, SciPy, Data Cleaning, Feature Engineering, Statistical Analysis, Window Functions

Tools & APIs: Matplotlib, Seaborn, Streamlit, REST APIs (python-binance), Git/GitHub, MS Excel, Jupyter, ANSYS Fluent, SolidWorks, MATLAB

PROJECTS

Predictive Maintenance System for Rotating Machinery

2026

Final Year Project · Group Leader, Team of 5 · Python, Scikit-learn, SciPy, Streamlit · github.com/Piyushsrivastava914/Predictive-Maintenance-System

- Led a team of 5 to design an end-to-end fault-classification pipeline for rotating-machinery vibration data, covering signal preprocessing and statistical feature extraction on the CWRU bearing dataset.
- Trained and tuned a Support Vector Machine (SVM) classifier that reached **92% classification accuracy** across multiple bearing fault states, benchmarked against baseline models.
- Deployed the model as an interactive Streamlit dashboard supporting waveform upload, FFT frequency-spectrum analysis, and real-time condition prediction.
- Documented methodology and results in a 19-slide technical presentation plus a structured README for reproducibility.

AI Mock Interview Platform

2026

Independent Project · Python, Streamlit, OpenAI API (GPT-4)

- Built an end-to-end interview-preparation platform that parses a candidate's resume and uses the OpenAI GPT-4 API to generate dynamic, role-specific technical questions across DSA, OOP, and system design.
- Implemented session persistence, solution validation, and a performance-analytics dashboard; owned the full product lifecycle from ideation and API design through deployment.

Automated Crypto Trading Bot — Binance Testnet

2025

Independent Project · Python, python-binance (REST API), Binance Testnet

- Built an automated trading bot on the Binance Testnet that combines multiple technical indicators to generate buy/sell signals and executes trades via the python-binance REST API.
- Implemented risk-management controls (position sizing, stop-loss logic) and structured logging of signals, orders, and account state to support strategy debugging and iteration.

EXPERIENCE

Intern, Advanced Manufacturing — IIT (BHU), Varanasi

Jun 2026 – Present

- Work on robotic Wire Arc Additive Manufacturing (WAAM) using the Cold Metal Transfer (CMT) process, applying data-driven methods to analyze deposition behavior and process parameters on an active robotic AM setup.

Intern — Bharat Heavy Electricals Limited (BHEL), Varanasi

May – Jun 2026

- Reviewed production workflow and quality-control data across industrial power-equipment manufacturing operations, applying engineering fundamentals to maintenance procedures.

EDUCATION

Madan Mohan Malaviya University of Technology, Gorakhpur

2023 – 2027

Bachelor of Technology, Mechanical Engineering | CGPA: 7.38 | Coursework: Data Analysis, Fluid Mechanics, Thermodynamics, Manufacturing Processes

CERTIFICATIONS

Data Analysis with AI — Internshala | Data Analysis — MOOD Indigo, IIT Bombay / Triple One Solutions